

SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

99.503: Nanoelectronics and Technology
Mid Term Test-2025

- Duration of the test: 2 hours
 - This is a closed book exam. No cell phones, notes nor books are allowed.
Cheat Sheet paper of size A4 written only on ONE Side is allowed.
 - The total marks of this test is **80 Marks**, which counts for **25%** of final grade.
 - A scientific calculator is allowed.
 - Write **clearly in pen [not pencil]**.
 - Remember to draw clear schematics/diagrams if necessary.
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****ALL THE BEST****

Part A

Q1. State whether the below statements are True or False

(2 marks)

- a) MOSFET is a current controlled device
- b) MOSFET shows low resistance path from drain to source under saturation condition
- c) p-channel MOSFET has a p-type doped source and drain
- d) Threshold voltage of a MOSFET is a function of gate dielectric thickness

Q2. What is the main purpose of doping intrinsic silicon?

(1 mark)

- a) To increase its atomic mass
- b) To improve its optical transparency
- c) To increase its electrical conductivity
- d) To change its crystal structure

Q3. When a trivalent impurity is added to silicon, it creates:

(1 mark)

- a) Excess electrons in the conduction band
- b) Holes in the valence band
- c) No change in carrier concentration
- d) Both electrons and holes equally

Q4. What type of impurity is added to create n-type silicon?

(1 mark)

- a) Trivalent impurity
- b) Pentavalent impurity
- c) Tetravalent impurity
- d) Hexavalent impurity

Q5. The concentration of intrinsic carriers (n_i) in silicon depends strongly on:

(2 mark)

- a) Pressure
- b) Temperature
- c) Crystal orientation
- d) Impurity type

Q6. What happens to the Fermi level when silicon is doped with a pentavalent impurity?

(1 mark)

- a) It moves closer to the conduction band
- b) It moves closer to the valence band
- c) It stays at the mid-gap
- d) It disappears

(2 marks)

Q7. State whether the below statements are True or False

- a) Implant energy in ion implantation primarily determines the penetration depth of ions.
- b) Better control over dopant depth and dose is an advantage of diffusion over ion implantation.
- c) During diffusion, the driving force for dopant movement is electric field.
- d) Diffusion is less preferred in modern CMOS fabrication compared to ion implantation due to poor dopant uniformity and depth control.

(2 marks)

Q8. State whether the below statements are True or False

- a) Film thickness and step coverage in Physical vapour deposition (PVD) are generally non-uniform on high-aspect-ratio features.
- b) PVD involves a chemical reaction between precursor gases to form a film.
- c) In Chemical vapour deposition (CVD), higher temperatures generally increase reaction rate and film growth rate.
- d) ALD has higher deposition rate compared to CVD.

(2.5 marks)

Q9. State whether the below statements are True or False

- a) Wet etching generally provides excellent anisotropy and precise pattern control.
- b) Wet etching can cause undercutting beneath the mask layer due to its isotropic nature.
- c) Dry etching is generally anisotropic, giving better directionality and feature definition.
- d) Dry etching is preferred for submicron and nanometer-scale pattern transfer.
- e) In reactive ion etching (RIE), only chemical reactions contribute to material removal.

(1 mark)

Q10. What causes the hot carrier effect in MOSFETs?

- a) Excessive gate voltage
- b) High electric field near the drain region
- c) Low threshold voltage
- d) Poor channel mobility

(1 mark)

Q11. In a CMOS inverter, noise margin represents:

- a) The gain of the inverter at the switching point
- b) The maximum noise voltage that can be tolerated without logic error
- c) The difference between input and output voltages
- d) The threshold voltage of the MOSFET

Q12. Which of the following statement/statements is/are correct with respect to e-beam patterning.

(2 marks)

- a) Can be used for mass/volume fabrication

- b) Masks are not required
- c) Slow process
- d) High resolution customised pattern with critical dimension-50nm

Q13. Which of the following materials is commonly used for interconnects in modern ICs?

- a) Silicon
- b) Copper
- c) Aluminum
- d) Both b and c

(1 Mark)

Q14. Low-k dielectrics are used in interconnect layers to:

- a) Reduce leakage current
- b) Reduce parasitic capacitance
- c) Increase power dissipation
- d) Improve mechanical strength

(1 Mark)

Q15. As the frequency of an AC signal increases, the skin depth:

- a) Increases
- b) Decreases
- c) Remains constant
- d) Becomes independent of material properties

(1 Mark)

Q12. The skin effect occurs due to:

- a) Resistance decreasing with frequency
- b) Induced eddy currents opposing deeper penetration
- c) Increase in the permittivity of the material
- d) Thermal effects at high frequency

(1 Mark)

Q13. State whether the below statements are True or False

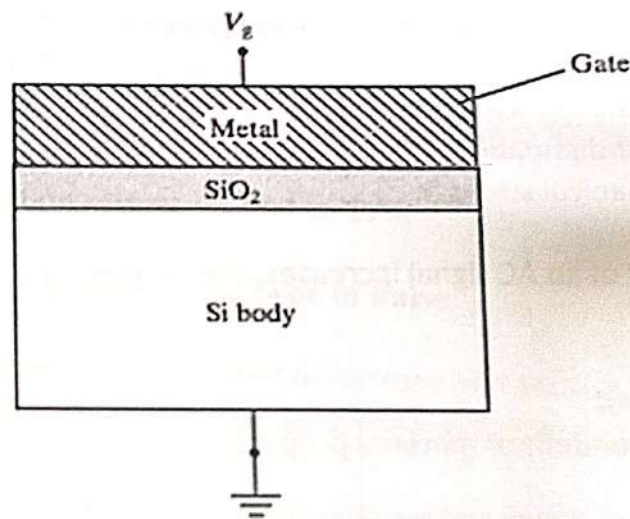
(1.5 Marks)

- a) Using a high-k dielectric allows a physically thicker gate insulator while maintaining high capacitance.
- b) High-k materials always have a higher bandgap than SiO_2 .
- c) The use of high-k dielectrics increases the equivalent oxide thickness (EOT).

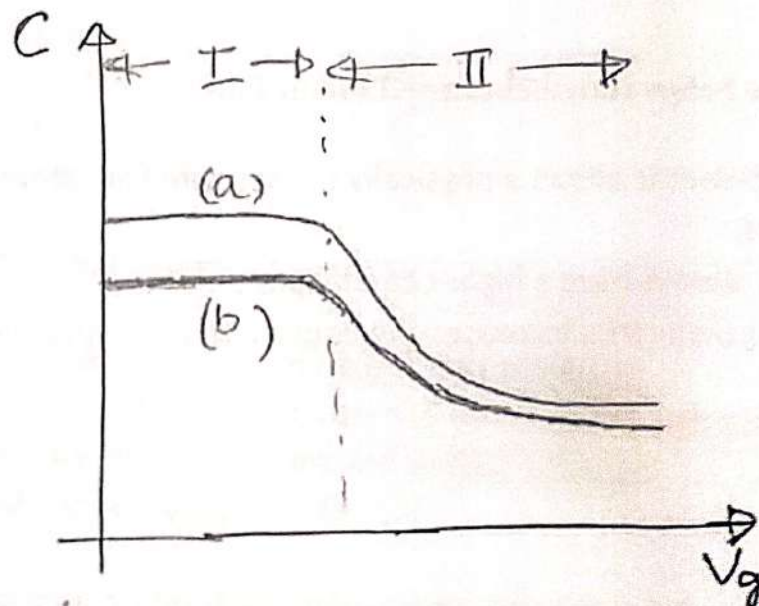
Part B

Q1. Briefly explain the Moore's law in MOSFET Scaling and highlight the key advantages of MOSFET scaling in silicon industry. (3 marks)

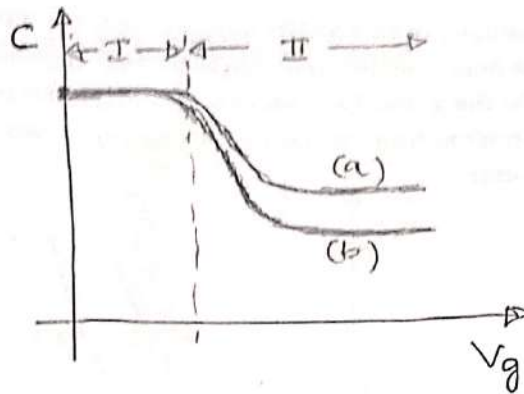
Q2. Ignoring the capacitance of the Silicon substrate, draw the equivalent capacitance network of the MOS capacitor shown below under inversion mode operated at high frequency. (2 marks)



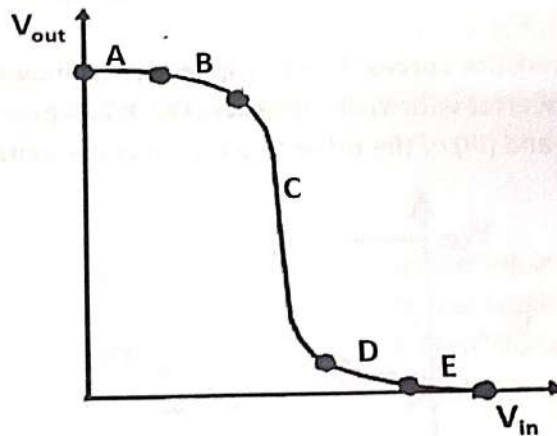
Q3. (a) Consider C-V graph (high frequency) of two MOS capacitors [(a) and (b)] as shown in figure below. Explain the reason for the shift in graph (b) compared to graph (a) only for the section (I). (2 marks)



(b) Consider C-V graph (high frequency) of two MOS capacitors [(a) and (b)] as shown in figure below. Explain the reason for the shift in graph (b) compared to graph (a) only for the section (II). (2 marks)

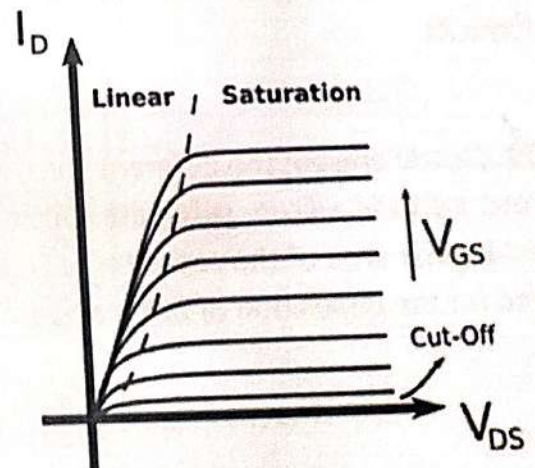
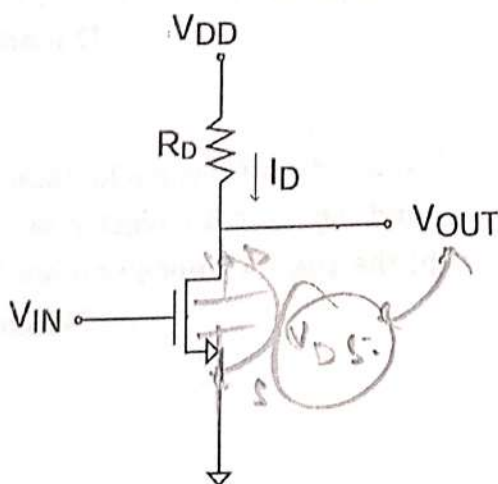


Q4. Draw the schematic of CMOS inverter with appropriate labels. Using the below transfer characteristics curve, indicate the different operating regions of NMOS and PMOS on the curve. (3 marks)



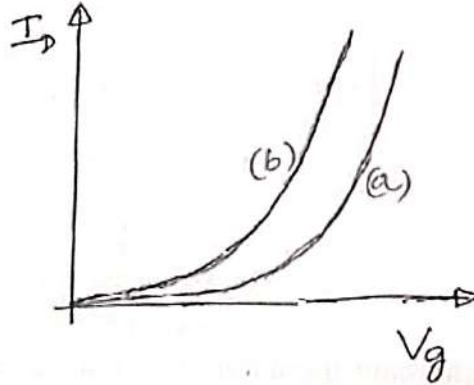
Q5 (a) Draw the DC loadline on the I_D - V_{DS} characteristics graph for the below 1T+1R NMOS inverter for $R_D=1K\Omega$ and $R_D=10K\Omega$. Ideal NMOS with $V_{DD}=5V$, $V_{in}=V_{GS}$ and $V_{out}=V_{DS}$

(b) Assume the NMOS has a leakage current of $0.5mA$ with $R_D=1K\Omega$ when $V_{GS}=0$. How does the DC loadline change compared to case (a) for $R_D=1K\Omega$. (4 Marks)

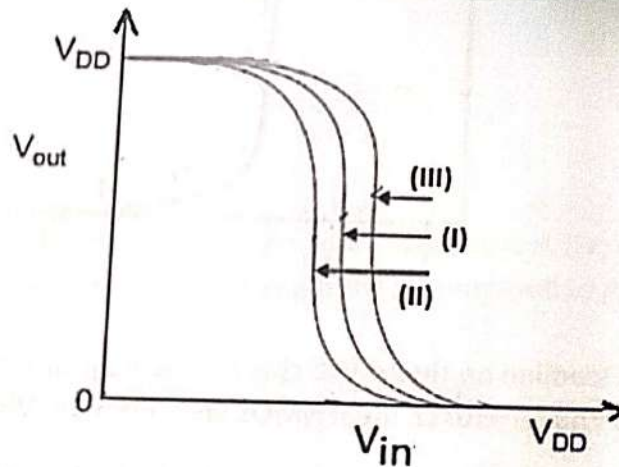


Q6. Briefly explain the velocity saturation effect in advanced MOSFETs. (2 marks)

Q7. (a) The I_D - V_g characteristics of two NMOS devices [device 1 and 2] is shown below. The NMOS devices have same doping levels, gate length and width. What parameter of MOSFET responsible for the shift in the graph from [device 1] (a) to [device 2] (b). Highlight the key parameter that can be extracted from the curves (a) and (b) of two respective MOSFETs and compare extracted parameter. (2 marks)



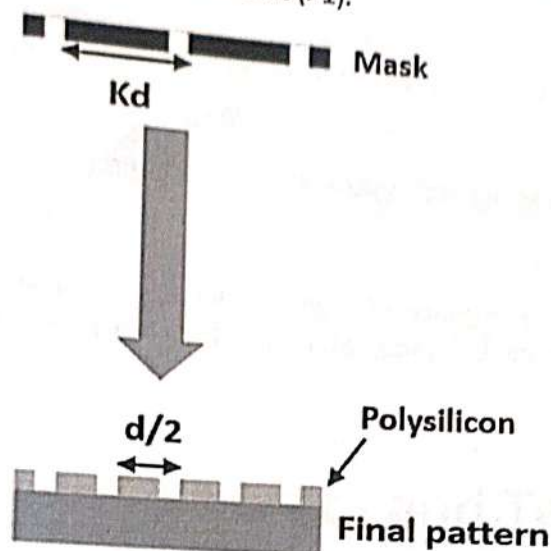
(b) The transfer characteristics curves 3 CMOS inverters is shown below. The curve (I) is obtained from the first inverter with $V_{th}(NMOS)=V_{th}(PMOS)$. Explain the possible reasons for the shift in the curves (II) and (III) of the other two inverters compared to curve (I). (2 marks)



Q8. Explain the impact drain-induced barrier lowering (DIBL) effect in MOSFET having a gate length L with the help of energy band diagram along the gate length for $V_{DS} = 0$, $(V_{dd}/2)$, and V_{dd} . (3 marks)

Q9. Clearly present the different steps involved in the fabrication of a MOS capacitor starting from intrinsic silicon substrate, with 10nm thick oxide and Ag as gate electrode. The rectangular area of the capacitor is $50\mu m \times 50\mu m$. Highlight the specific deposition method used for the deposition of oxide and gate electrode. (4 marks)

Q10. It is required to fabricate the final polysilicon pattern on Silicon substrate as shown in figure below. Sketch the detailed steps used during the whole lithography process. Optical pitch of the mask is Kd , where K is a constant (>1).

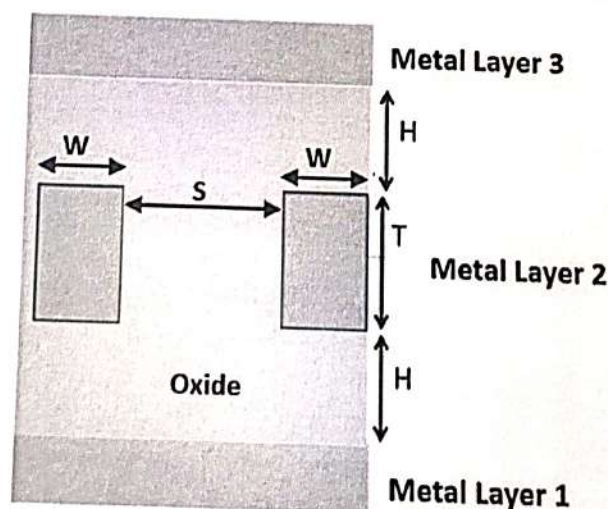


(4 marks)

Q11. Discuss the I_{on} , I_{off} with help of graphs and Electric field distribution in FinFETs (Tri-gate) compared to bulk MOSFET. (3 marks)

Q12. Briefly explain the working principle of e-beam sputtering and Atomic Layer Deposition (ALD) technique for the deposition of films/metals. List few applications where e-beam sputtering and ALD are relevant techniques to deposit films/metals. (4 marks)

Q13. Indicate Resistance components separately for the interconnect wires shown below. For (Metal layer) Wire 1 and 3, resistivity is ρ_1 , cross sectional area A_1 and length L_1 . For (Metal layer) Wire 2, resistivity of wire 2 is ρ_2 , cross sectional area A_2 and length L_2 . Indicate and express the capacitance network between metal layer 1 and 3, with dielectric constant of the oxide (dielectric) is K .



(3 marks)

Q14. Briefly explain the electro-migration effect in interconnect conducting wires. Describe why electromigration effect becomes critical with interconnect scaling. (4 marks)

Q15. (a) Why do global interconnects experience higher RC delay compared to local interconnects?

(b) What are the techniques used to reduce RC delay in interconnect technology?

(4 marks)

Q16. Briefly explain the significance of High-K dielectrics in Advanced CMOS technology. Highlight the major challenges in integrating high-k dielectric material advanced MOSFET structure.

(5 marks)